

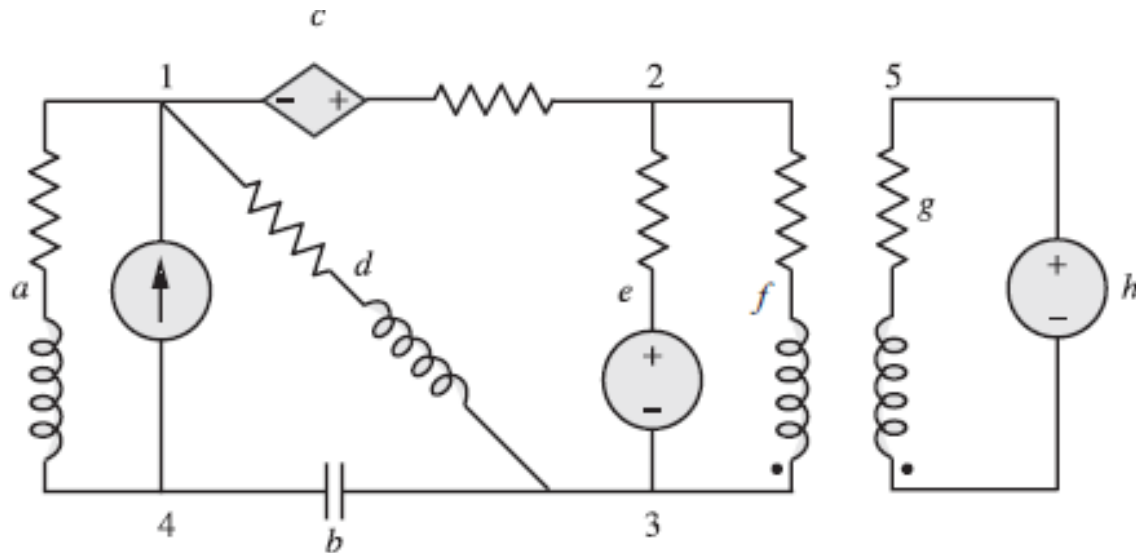
Module -5

Network Graphs

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SENSE

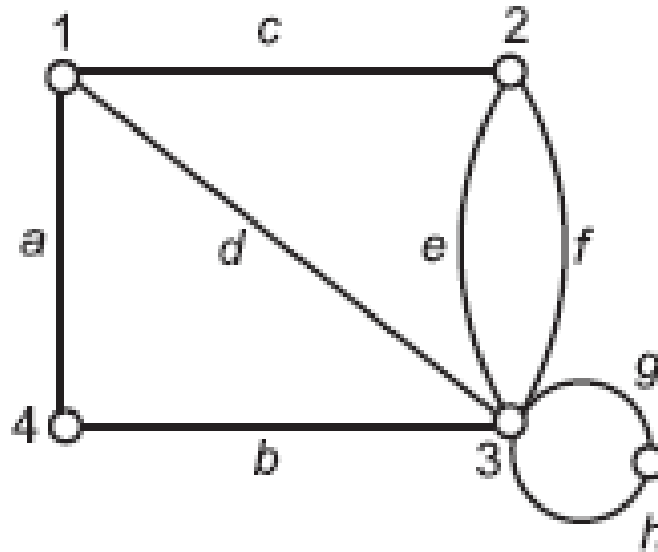
Definition of Basic Terms

- **Network:** A combination of two or more network elements is called a network.
- **Circuit:** If the network has atleast one closed path, it is a circuit.
Every circuit is a network but every network is not a circuit.



- **Graph:**

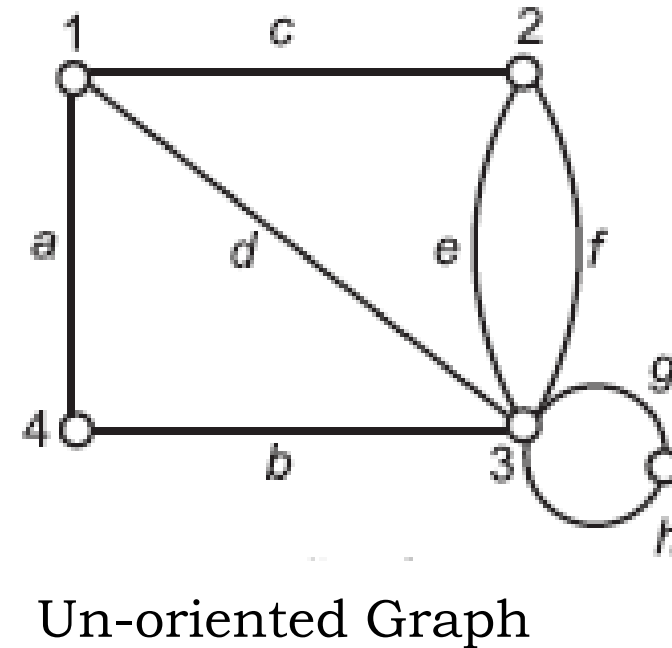
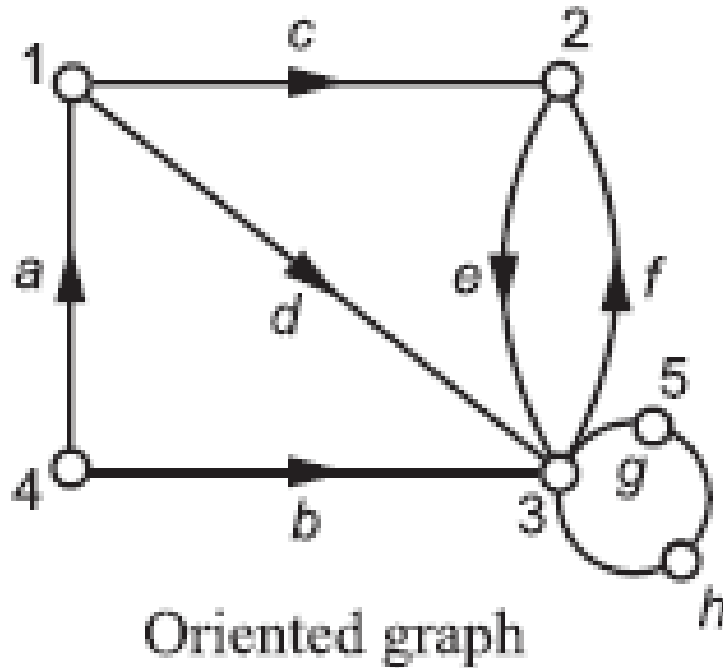
- It is a network in which all nodes and loops are retained but its branches are represented by lines.
- Voltage sources are replaced by short circuits and current sources are replaced by open circuits.



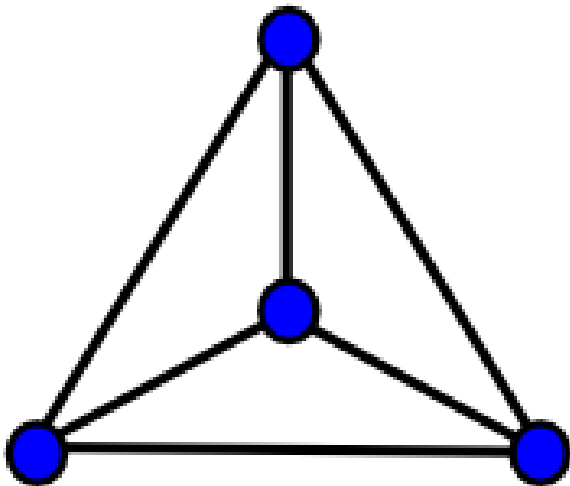
Graph

- **Branch:** A line segment replacing one or more network elements that are connected in series or parallel.
- **Node:** Interconnection of two or more branches. It is a terminal of a branch. Usually interconnections of three or more branches are nodes.
- **Degree of Node:** Number of branches incident on the Node is the degree of the Node.
- **Path:** A set of branches that may be traversed in an order without passing through the same node more than once.
- **Loop:** Any closed contour selected in a graph.
- **Mesh:** A loop which does not contain any other loop within it.

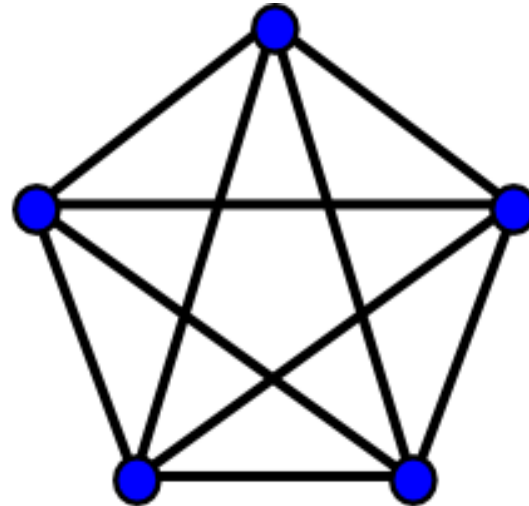
- **Oriented graph:** When a direction to each branch of a graph is assigned, the resulting graph is called an oriented graph or a directed graph.
- **Un-Oriented graph:** A graph without orientation of the branch.



- **Planar graph:** A graph which may be drawn on a plane surface in such a way that no branch passes over any other branch.
- **Non-planar graph:** Any graph which is not planar.

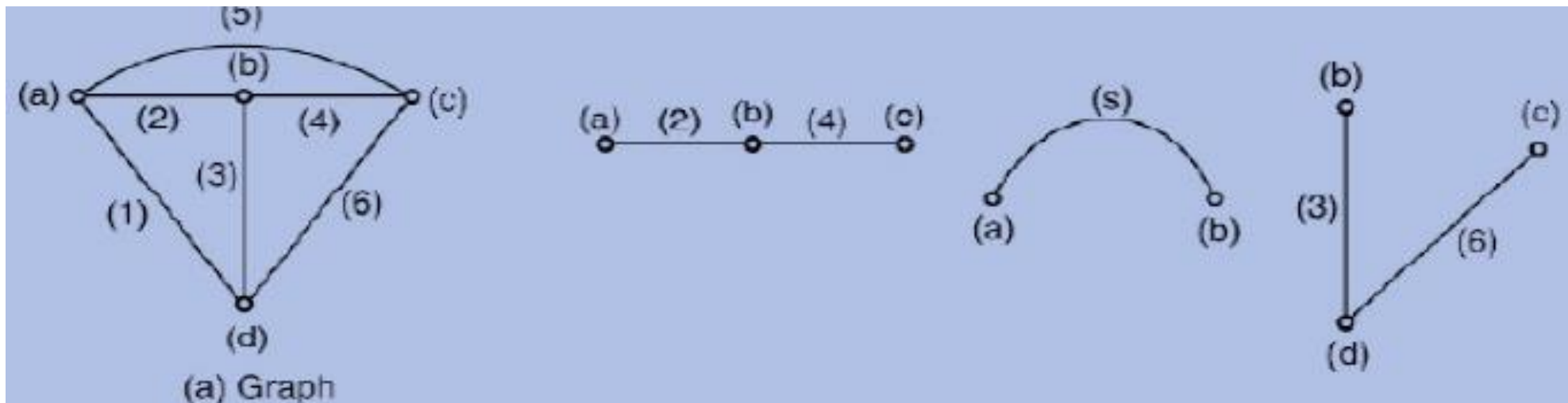


Planar Graph

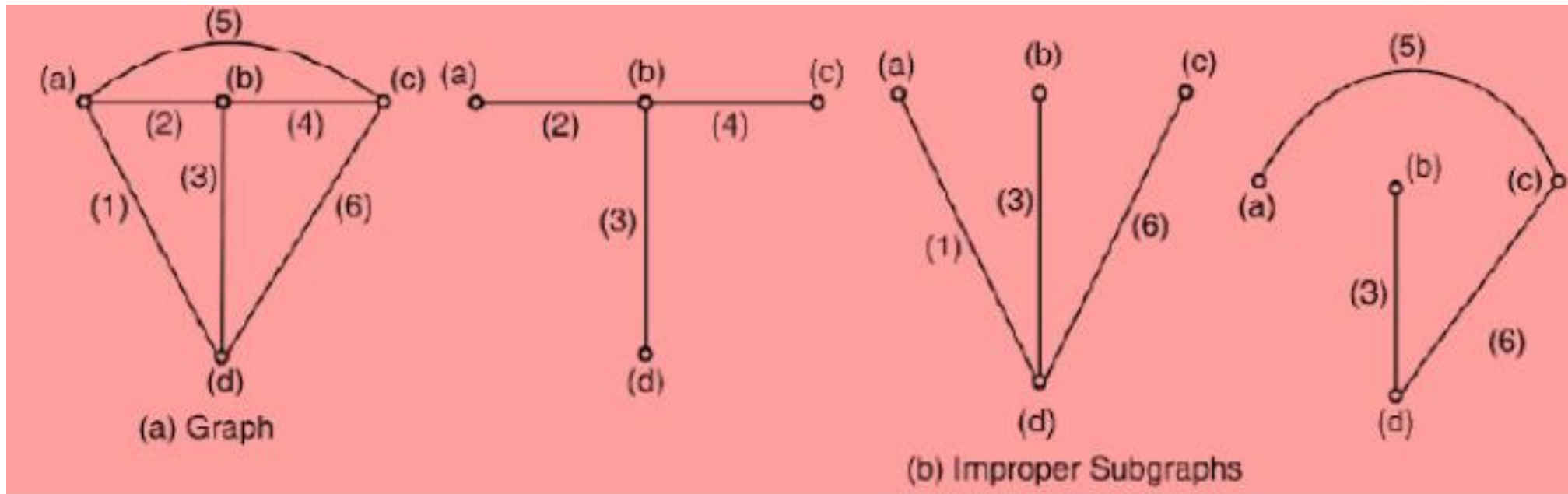


Non-Planar Graph

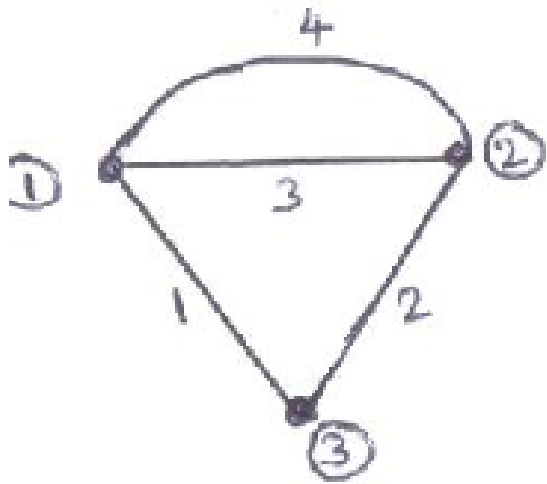
- **Sub graph:** Any subset of branches of the graph.
- **Proper Sub graph:** If the graph contains branches and nodes less in number than those on the Graph, then such a subgraph is called proper sub graph.



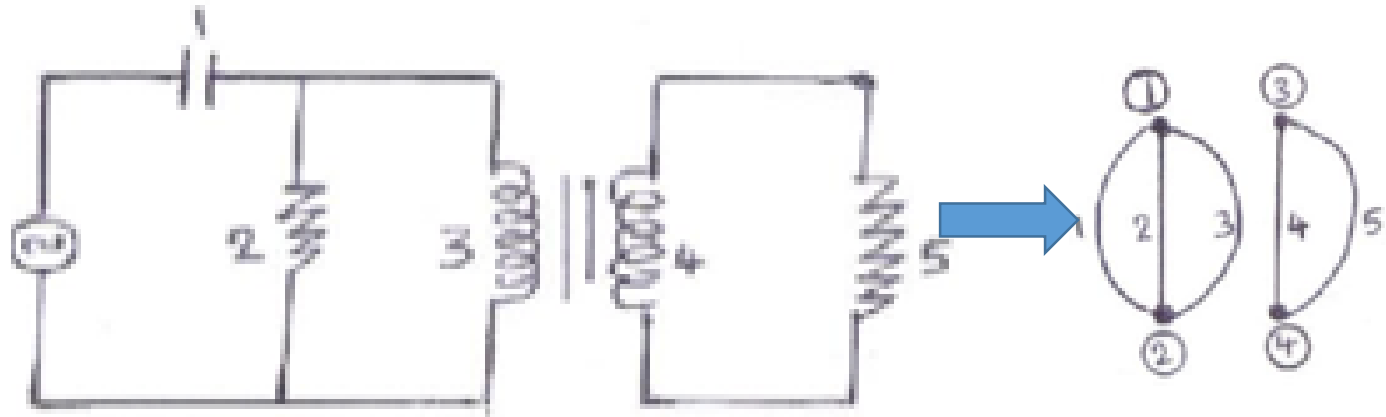
- **Improper Sub graph:** If the sub graph contains all the nodes and some of the branches of the graph, then such a subgraph is called improper sub graph.



- **Connected graph:** A graph is connected if and only if there is a path between every pair of nodes.
- **Un-connected Graph:** If there exists no path from each node to every other node, the graph is said to be un-connected graph.

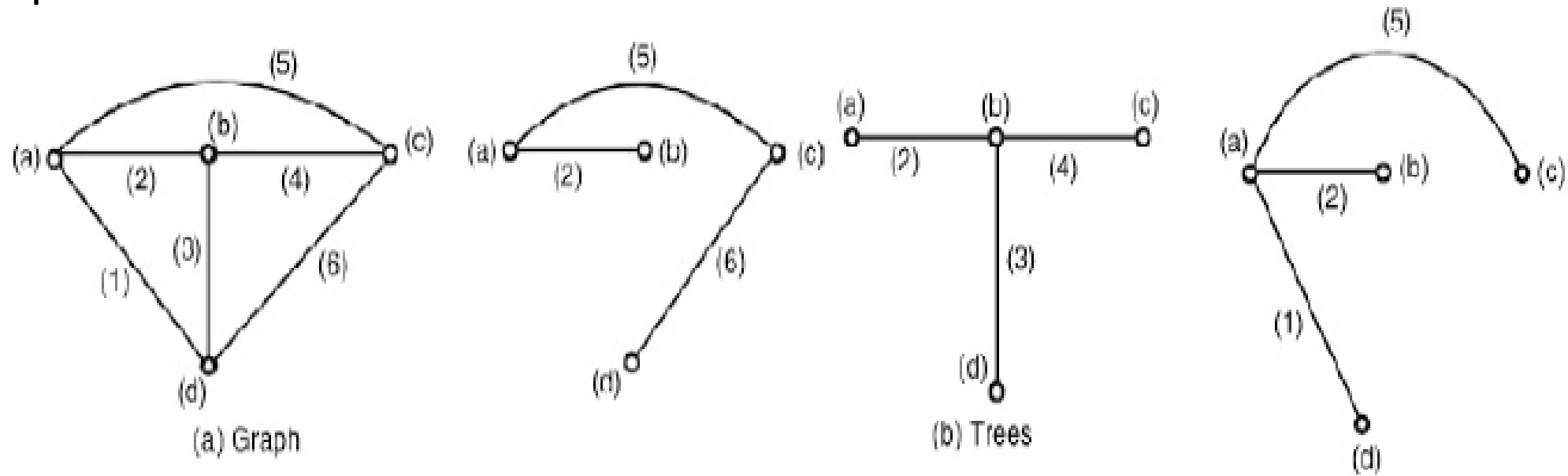


Connected Graph



Unconnected Graph

- **Tree:** It is a connected sub-graph containing all nodes of a graph but no closed path. It is a set of branches of graph which contains no loop but connects every node to every other node not necessarily directly. A number of different trees can be drawn for a given graph.



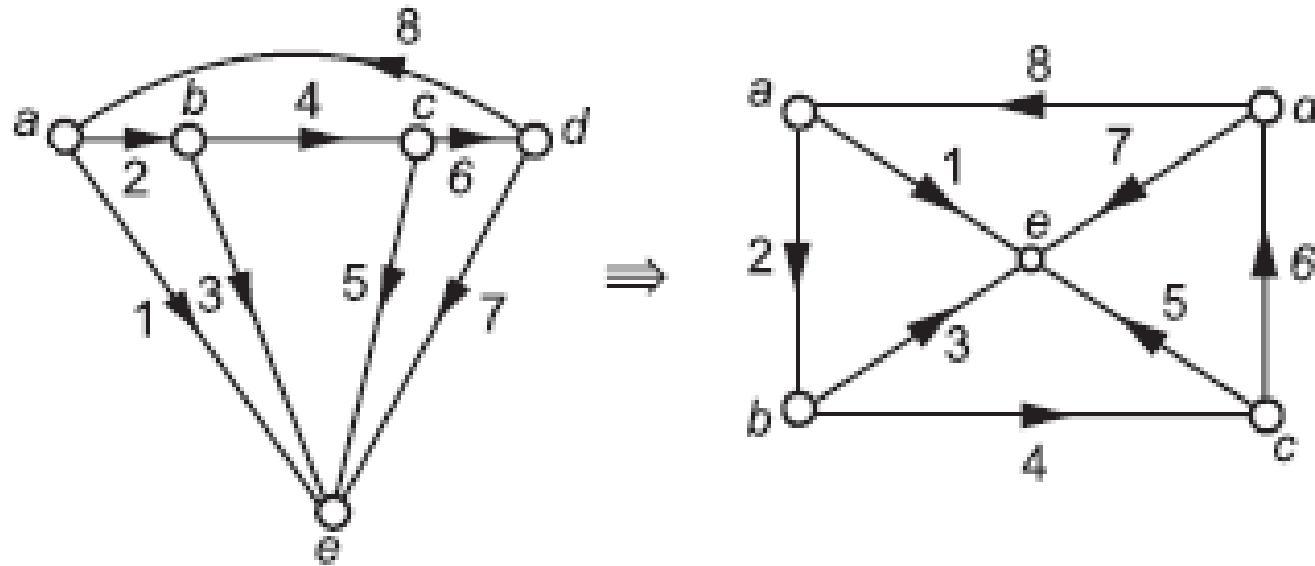
- **Properties of Tree:**

- Tree contains all the nodes of the graph
- Tree does not contain any closed path or loop
- In a tree, there exists only one path between any pair of nodes
- Every connected graph has at least one tree.
- The number of terminal nodes of every tree are two.
- A connected subgraph of a connected graph is a tree if there exists all the nodes of the graph.
- Each tree has $(n-1)$ branches, where n is the number of nodes of the tree
- The rank of a tree is same as the rank of graph, $(n-1)$

- **Twigs:** The branches of a tree are called its 'twigs'.
- **Co-tree:** The complementary set of branches of the tree is called the co-tree of the graph.
- **Links:** The branches of co-tree are called links.
- **Relation between nodes, links, and branches:**
 - Let B = Total number of branches in the graph or network
 - N = total nodes, L = link branches
 - Therefore number of independent node pair voltages = $N - 1$ = number of tree branches.
 - Then $L = B - (N - 1) = B - N + 1$
 - Number of independent loops = $B - N + 1$

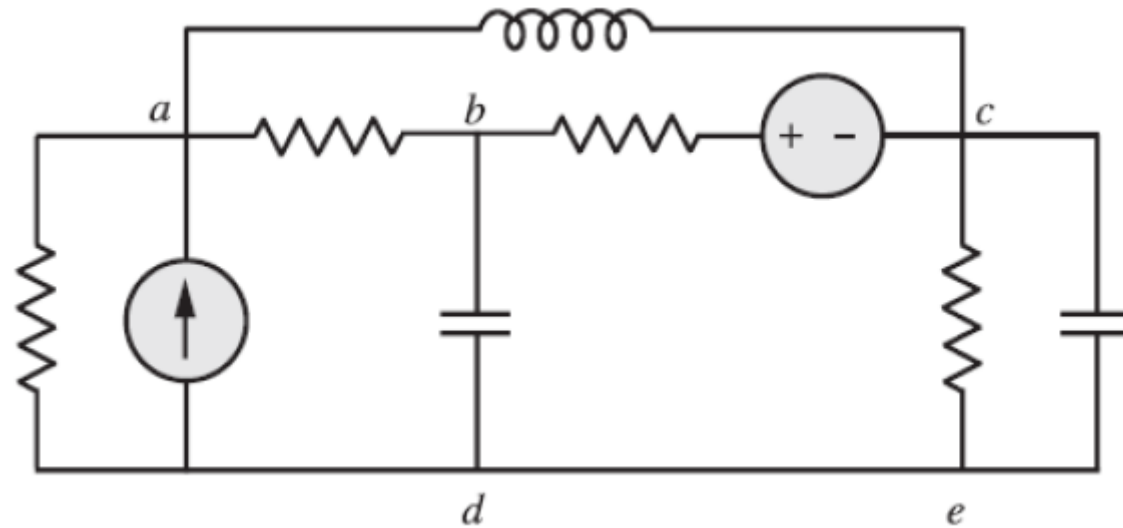
- **Isomorphic graphs:**

Two graphs are said to be isomorphic if they have the same incidence matrix, though they look different. It means that they have the same numbers of nodes and the same numbers of branches. There is one to one correspondence between the nodes and one to one correspondence between the branches.

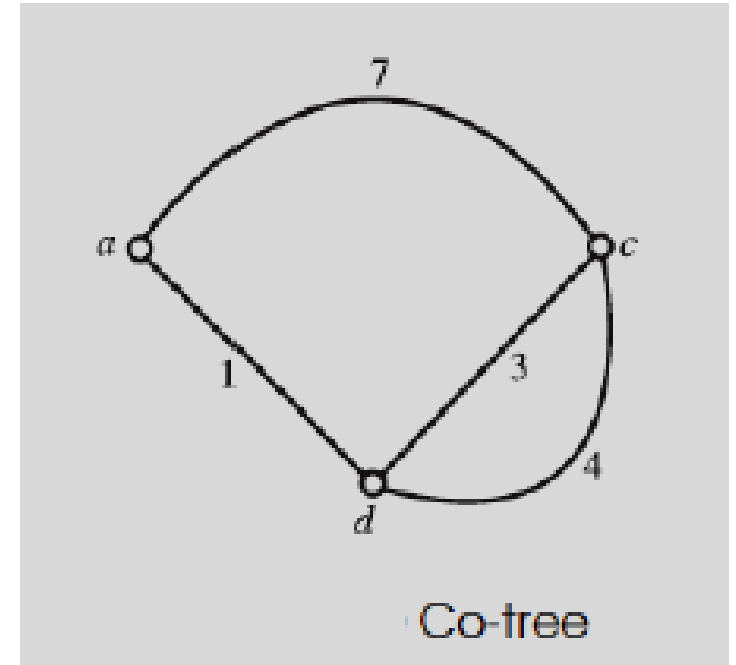
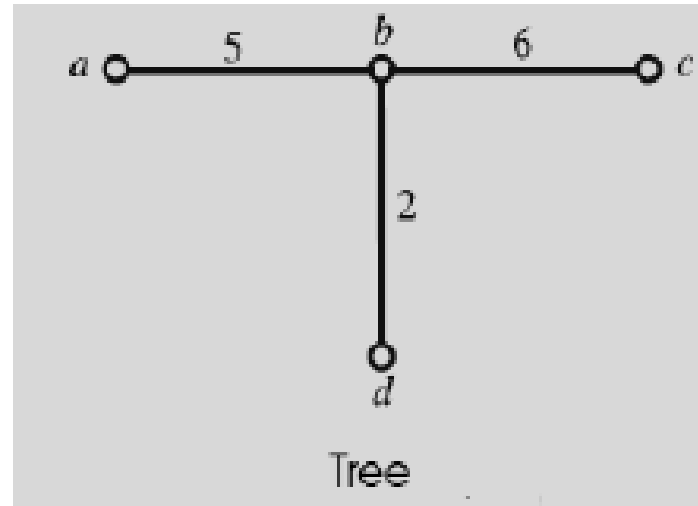
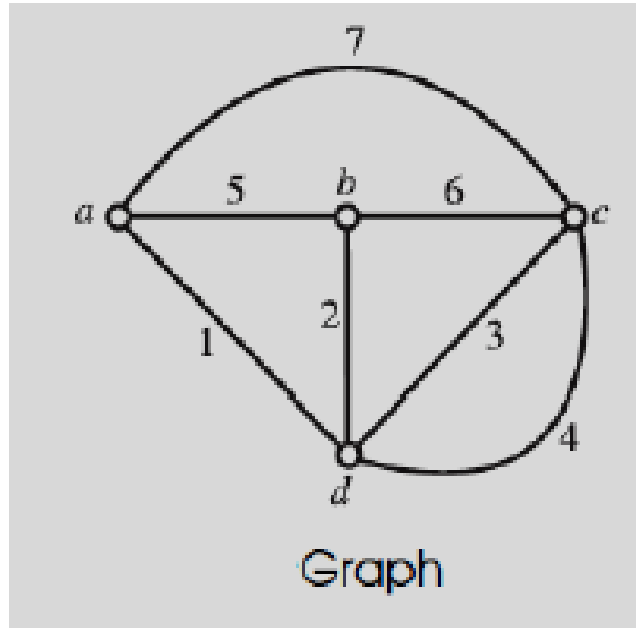


Numerical - 1

- Draw the graph, tree and its co-tree.

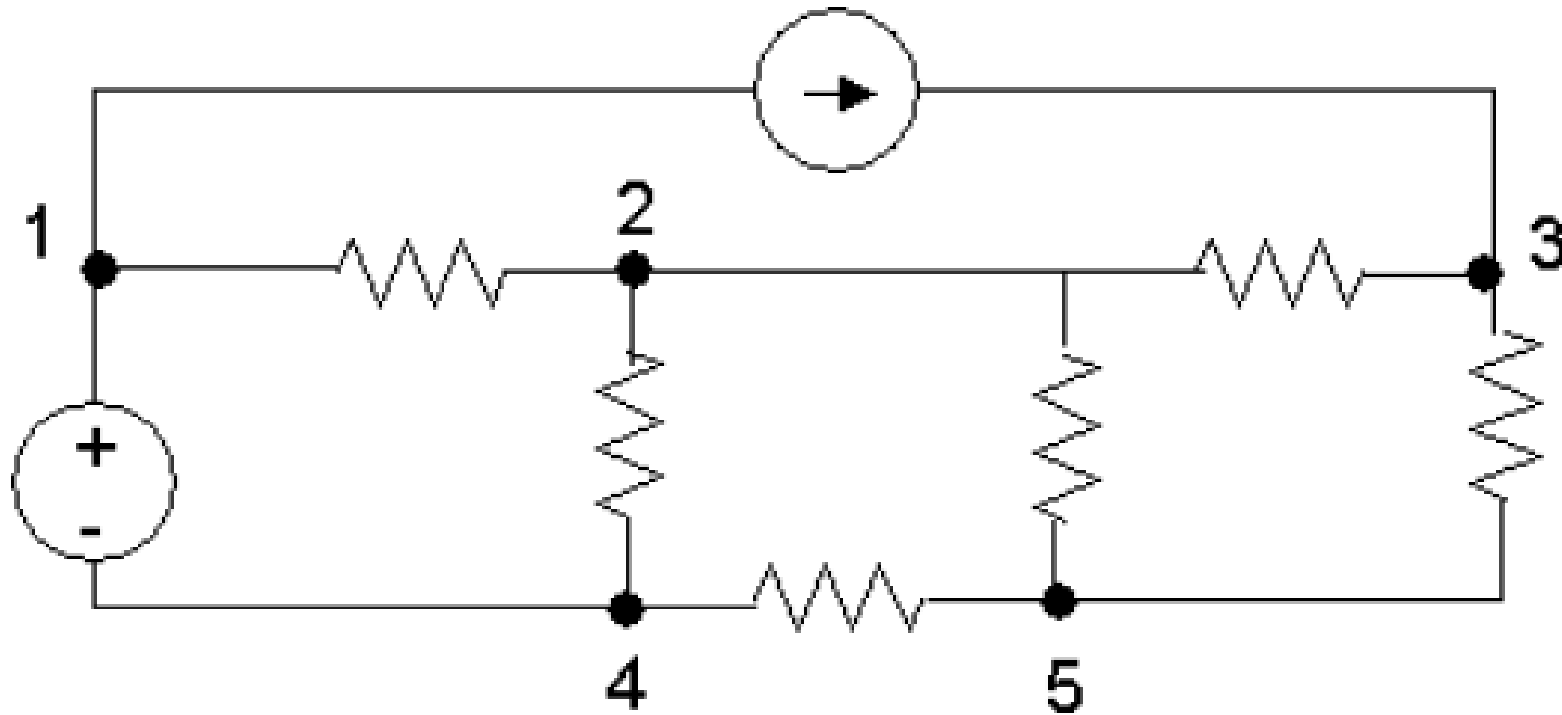


Solution:

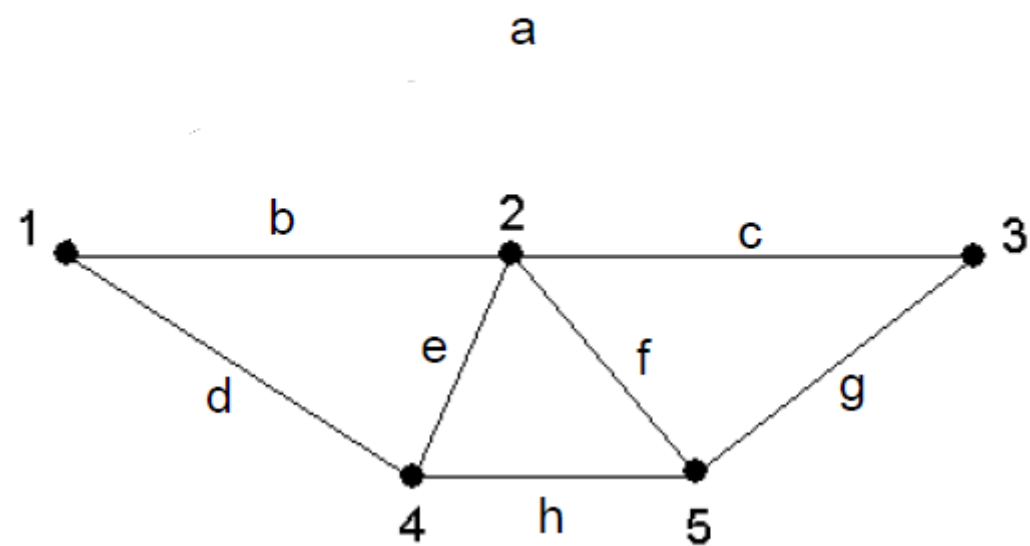


Numerical - 2

- Draw the graph, tree and its co-tree.



Graph of a Circuit



Trees and Co-Trees

